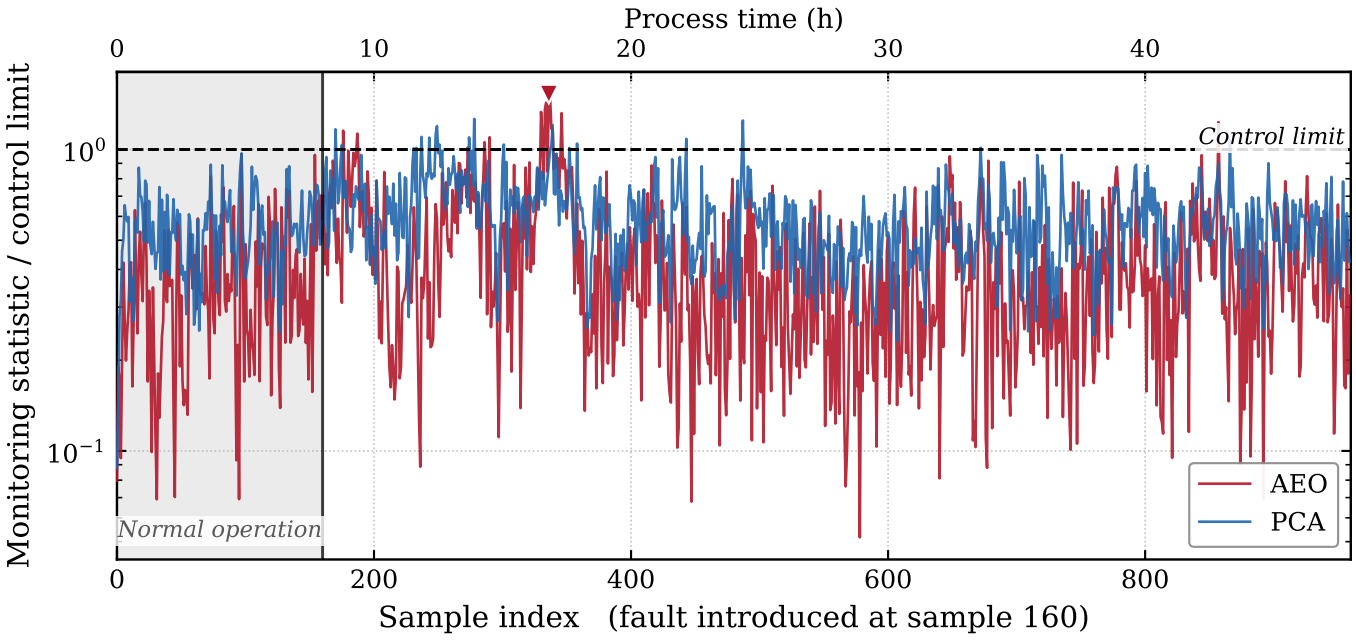
Fault Detection Performance for IDV(15)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay 221.0 samples (663 min)	MDR 0.987	detected in 2/7 runs
PCA	delay 337.0 samples (1011 min)	MDR 0.987	detected in 1/7 runs

Fault not reliably detected by either method (caught in fewer than half the runs)